

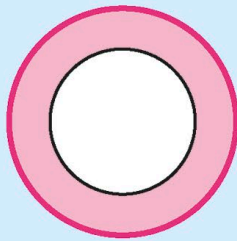
Drawing a face

A cartoonist draws a picture of a friend's face. Another friend sees it and asks for a cartoon of her own face. There are common features and shared steps that can help the artist to come up with the second illustration quickly and efficiently. The pattern from the first drawing can be used as many times as needed.

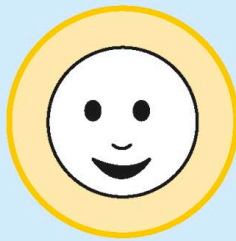
▽ The drawing process

The pattern of steps involved in drawing each cartoon is almost identical, but the outcome is quite different. In fact, the only difference is the hairstyle – spiky in one cartoon and straight in the other.

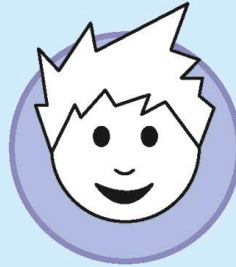
Drawing a face with spiky hair



Draw a circle



Add eyes, mouth, and nose

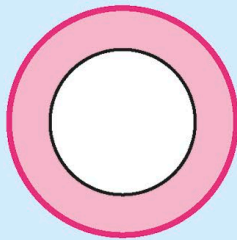


Add spiky hair

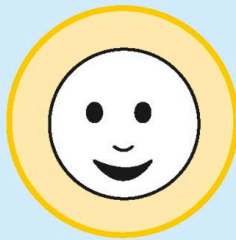


Colour in

Drawing a face with straight hair



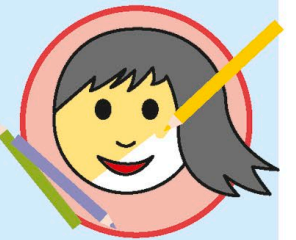
Draw a circle



Add eyes, mouth, and nose



Add straight hair



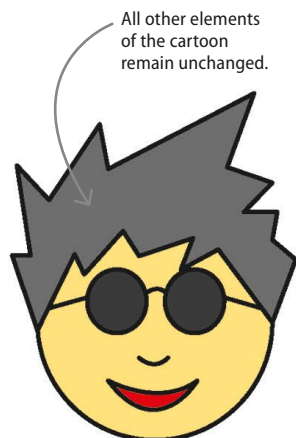
Colour in

Reusing old solutions

If a new problem, or elements of a new problem, are similar to a problem that has been fixed before, it is possible to reuse the old solution to fix the new problem. Recognizing patterns saves time as it cuts down on the effort needed to fix a problem.

▷ Altering a pattern

To draw the first face (above) with sunglasses, a cartoonist can take the solution used above and add one extra step. Once again, a completely different outcome is possible by slightly altering a pattern.



REAL WORLD

Success and failure

The HOLMES 2 computer system helps the UK police solve crimes by matching patterns of criminal behaviour. Pattern recognition doesn't always lead to solutions, however. An unsuccessful initiative involving patterns in data was Google's Flu Trends program. It hoped to identify outbreaks of flu from clusters of flu-related searches in particular parts of the world. However, the high amount of search data didn't correspond with the outbreaks.